Hong-Liang LIN

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Github: https://hong-liang-de.github.io/

Education

2024 – now	**M. Sc. Mechanical Engineering** Technical University of Berlin, Germany
2018 – 2022	**B.Sc. Mechanical and Energy Engineering** National Chiayi University, Taiwan

Internship & Experiences

07.2026 – present
Individual Project. Lightweight Optimization of a Composite UAV Rotor Arm, Dept. Aircraft Design and Aerostructures, TU Berlin
Task: Development of a Python-based laminate analysis and optimization tool using Classical Laminate Theory (CLT), combined with FEM-based structural validation of a CFRP sandwich UAV rotor arm.

05.2026 – present
Student Research Assistant. Federal Institute for Materials Research and Testing (BAM) - Dept. 8.5, Berlin, Germany
Task: Redesign and optimisation of a high-temperature tensile testing system, in-situ X-ray CT analysis, and correlation of experimental observations with FEM models for material characterisation and structural prediction.

04.2026 – 07.2026
Project. Machine Learning for Material Modeling of Cyclic Plasticity, Dept. Structural and Computational Mechanics, TU Berlin
Task: Development of neural-network-based constitutive models for cyclic plasticity using experimental stress–strain data.

04.2025 – 04.2026
Project. FFKu - Ballastless Track Constructed from Recycled Polymer Materials, Dept. Structural and Computational Mechanics, TU Berlin
Task: Research on the use of recycled plastics in ballastless track systems to evaluate the practical application in the future. This project is funded by the Federal Ministry for Economic Affairs and Climate Action (BMWK).

09.2022 – 11.2022
Internship. Metal Industries Research & Development Centre, Taiwan.
Task: Non-destructive testing and welding to develop offshore wind power plant technology.

07.2020 – 07.2021
Project. Fatigue and lightweight design development of aluminum die-cast wheel rims for electric motorcycles.
Task: Determination of the fatigue properties of YAMAHA-certified die-casting mateiral YDC-19 and analysis of the fatigue life of wheel rims using Altair HyperWorks and Inspire.

Research

2021 Hong-Liang, Lin, Chia-Chin, Wu*, Structural Strength Analysis of Motorcycle Swingarm, The 37th National Conference on Mechanical Engineering of CSME

Skills

Languages
Native: Mandarin Chinese
Fluent: English (C1), German (C1)
Intermediate: Korean (B1)

Software
CAD: Autodesk Inventor, Solid Edge
FEM: Altair HyperWorks, Altair Inspire, ABAQUS, ANSYS
Programming: Python, MATLAB